

main.py code:

```
from fastapi import FastAPI, HTTPException, Depends
from fastapi.security import (
    OAuth2PasswordBearer,
    OAuth2PasswordRequestForm )

from jose import jwt
app = FastAPI()
SECRET_KEY = "my-secret-key"
ALGORITHM = "HS256"
oauth2_scheme = OAuth2PasswordBearer(
    tokenUrl="login"
)

@app.post("/login")
def login(
    form_data: OAuth2PasswordRequestForm = Depends()
):
    if (
        form_data.username == "Charan"
        and form_data.password == "Charan123"
    ):
        payload = {
            "sub": form_data.username,
            "role": "student"
        }
        token = jwt.encode(
            payload,
            SECRET_KEY,
            algorithm=ALGORITHM
        )
        return {
```

```

        "access_token": token,
        "token_type": "bearer"
    }
    raise HTTPException(
        status_code=401,
        detail="Invalid username or password"
    )

def get_current_user(
    token: str = Depends(oauth2_scheme)
):
    try:
        payload = jwt.decode(
            token,
            SECRET_KEY,
            algorithms=[ALGORITHM]
        )
        return payload
    except Exception:
        raise
        HTTPException(
            status_code=401,
            detail="Invalid or expired token"
        )

@app.get("/profile")
def profile(
    user=Depends(get_current_user)
):
    return {
        "message": "Welcome to your profile",
        "user_id": user["sub"],
        "role": user["role"]
    }

```

POST

POST /login Login

Parameters

No parameters

Request body required

grant_type
string | (string | null)
pattern: ^password\$

password

☐ Send empty value

username * required
string

Charan

password * required
string(\$password)

scope
string

scope

☒ Send empty value

client_id
string | (string | null)

string

☐ Send empty value

client_secret
string | (string | null){\$password}

☐ Send empty value

Execute

Response:

Responses

Curl

```
curl -X 'POST' \
  http://127.0.0.1:8000/login \
  -H 'accept: application/json' \
  -H 'Content-Type: application/x-www-form-urlencoded' \
  -d 'grant_type=password&username=Charan123&word=priyanka123&scope=&client_id=string&client_secret=*****'
```

Request URL

http://127.0.0.1:8000/login

Server response

Code	Details
200	Response body <pre>{ "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ2ZWdT10lQcm15YW50YSIsInR5bGU10l3ZdMv26C0ln0. sgVUoqQUVvJo4VbvX_x_-7k2c10dFBm8qk1DnJ00knc", "token_type": "bearer" }</pre> Response headers <pre>content-length: 169 content-type: application/json date: Tue, 15 Sep 2026 05:35:18 GMT server: uvicorn</pre>

Authorization:

Available authorizations x

Scopes are used to grant an application different levels of access to data on behalf of the end user. Each API may declare one or more scopes.

API requires the following scopes. Select which ones you want to grant to Swagger UI.

OAuth2PasswordBearer (OAuth2, password)

Authorized

Token URL: login
Flow: password
username: Charan
password: *****
Client credentials location: basic
client_secret: *****

LogoutClose

GET:

GET /profile Profile  

Parameters Cancel

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/profile' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJQcm15Ym5rYSIsInVubGUzIjpm681nd. sgVloqQUVvYj04VbVX_x_-7k2c1D0F8m8q1DnJ00knc'
```

Request URL

http://127.0.0.1:8000/profile

Server response

Code

Details

200

Response body

```
{
  "message": "Welcome to your profile",
  "user_id": "Charan",
  "role": "student"
}
```

 Download

Response headers

```
content-length: 75
content-type: application/json
date: Tue, 15 Sep 2026 05:48:14 GMT
server: uvicorn
```